

```
$ mysql -e "SHOW CREATE TABLE orders" --normalize
```



// 1NF → 2NF → 3NF □□□□□

1NF

□□□

2NF

□□□□□□

3NF

□□□□□□□

□□□□

OLAP □□□

\$ cat schema-plan.md



01 / 
□□□□□□□□

02 / 
1NF    → 2NF → 3NF

03 / 
JOIN □□□

04 / 
OLTP vs OLAP □□□

□□□□□□□□□□ — □□□□□□□□

PART 01



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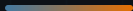
order_id	customer_name	customer_email	product_name	price
1	□□	<u>zhang@example.com</u>	□□□□□	30000
2	□□	<u>zhang@example.com</u>	□□	500
3	□□	<u>li@example.com</u>	□□	1200

□□□□

□□□□□□□□ email □□□□

□□□□

□□□ email□□ UPDATE □□□□



ROOT CAUSE



CHECKPOINT_01

データベースの UPDATE 37 回実行後 3 分経過

A データが更新された

B データが削除された

C データが追加された

PART 02



1NF → 2NF → 3NF →

1NF

NO DUPLICATES

BEFORE —

student	courses
	Maths, Maths, Maths, Maths, Maths, Maths, Maths

AFTER — 1NF

student	course
	Maths
	Maths
	Maths

2NF NOGRAPH

BEFORE — □□□□

order_id	product_id	product_name	qty
1	P1	□□	1
2	P1	□□	3

AFTER — □□□□□

order_id	product_id	qty
1	P1	1
product_id	product_name	
P1	□□	

3NF

NO SLIP

□□□□□□

student_id → dept_id → dept_name — dept_name — dept_id —

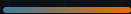
BEFORE — □□□□

student_id	name	dept_id	dept_name
S1	□□	D1	□□□
S2	□□	D1	□□□

AFTER — □□□□□

student_id	name	dept_id
S1	□□	D1
dept_id	dept_name	
D1	□□□	

□□□□□□□□□□ — □□□□□□□□□□



00000000000000000000 3NF 0000 0000000000000000

PART 03



□□□□□□JOIN □□□

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JOIN □□
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□□□□
JOIN □□□□□

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OLTP	OLTP	OLTP
OLTP	OLTP	OLTP
OLTP	OLTP	OLTP
OLTP	OLTP JOIN	OLTP JOIN
OLTP	OLTP	OLTP
OLTP	OLTP	OLAP



→



1NF                                               

→

3

1

1

□ □ □ □ □ □ □ □ □ □ □ □ □

OLTP □□□□OLAP □□□□□□□□

Schema normalized.

XXXXXXXXXXXX JOIN XXXXX

```
$ mysql -e "EXPLAIN SELECT ..."
```

1NF → 2NF → 3NF → XXXXXXXX